

Week 9

Aim

My aim for this lab was to be able to install and run local LLMs on my virtual machine and observe, assess and evaluate its behaviour. It also meant understanding how to attack models using prompt engineering skills.

Glossary

GenAI – AI systems that can produce novel content

LLM – A type of generative AI models trained on large text datasets to generate responses

Ollama – Open-source platform to run LLMs locally.

Models pull – The process of downloading the LLMs

Prompt injection – A security attack that crafts prompt to override the system instructions.

Data Poisoning – A threat where misleading or malicious data is intentionally fed into a model to change its behaviour

Model Inversion – A privacy attack where an adversary questions to model to reconstruct information from its training data.

Errors, Solutions and Reflections

During my time working on this seminar, I found difficulties in understanding the large language models initially because I didn't understand what was going on when they were installing on my computer. However, after inputting the code given from the worksheet, I was able to understand more about how I could utilise LLM locally on my machine. Another issue I came across was understanding how to format the LAB reflection in a way that was organised and I also found that a lot of the most models had similar responses except for one which was the smallest model out of the two. This made it hard to fully evaluate the models since they all had similar responses and therefore, I found my evaluation to be quite limited.

Additionally, an error I ran into was when I was installing one of the models, I accidentally installed one that was way too large which caused my VS code to crash. The way I overcame this challenge was I restarted my VS code and deleted the

instalment code for the model and replaced it with a much smaller model after I did a bit more research.

Overall, I felt as though this lab and lab 6 were my weakest lab because they involved system AI and involved deeper malware analysis which I found were my weak points as I did not have a lot of initial knowledge of these concepts. However, I found them quite interesting and I can confidently say I gained a deeper understand especially of LLMs since initially I did not really understand what they were until this lab.

Key takeaways

Data Poisoning - Understanding that you can feed a model misinformation which can later cause inaccuracy in responses and shows to us how important it is to secure datasets.

No model is completely safe - testing multiple models shows differences inconsistencies and susceptibility to attacks and therefore security must be evaluated in different models not just one-size-fits-all

Effective defence requires multiple layers - I learned as well from this lab that effective defences often require multiple layers of security not just one concept that covers all since there are such a wide variety of concepts to consider.